

Finite-Window Tilted Wald Second Moment for Erlang-2 Local Guess–Kill

1 Canonical Wald Moment

Let

$$X \sim \text{Wald}(a, v, \sigma), \quad a > 0, \quad \sigma > 0,$$

with density

$$f_X(x; a, v, \sigma) = \frac{a}{\sqrt{2\pi\sigma^2 x^3}} \exp \left[-\frac{(a - vx)^2}{2\sigma^2 x} \right].$$

For a tilt rate $r \geq 0$, define

$$q_r = \sqrt{v^2 + 2\sigma^2 r}, \quad L(r) = \exp \left[\frac{a}{\sigma^2} (v - q_r) \right].$$

Then

$$f_X(x; a, v, \sigma) e^{-rx} = L(r) f_X(x; a, q_r, \sigma).$$

The required second tilted raw moment is

$$M_2(U; r) = \int_0^U x^2 f_X(x; a, v, \sigma) e^{-rx} dx = L(r) I_2(U; q_r),$$

where

$$I_2(U; q) = \int_0^U x^2 f_X(x; a, q, \sigma) dx.$$

For finite $U > 0$, define

$$d_- = \frac{qU - a}{\sigma\sqrt{U}}, \quad d_+ = -\frac{qU + a}{\sigma\sqrt{U}}, \quad c = \frac{2aq}{\sigma^2},$$

$$A = \Phi(d_-), \quad B = e^c \Phi(d_+).$$

Then

$$I_0(U; q) = A + B, \quad I_1(U; q) = \frac{a}{q} (A - B),$$

and

$$I_2(U; q) = \left(\frac{a^2}{q^2} + \frac{a\sigma^2}{q^3} \right) A + \left(\frac{a^2}{q^2} - \frac{a\sigma^2}{q^3} \right) B - \frac{2a\sigma\sqrt{U}}{q^2} \phi(d_-).$$

Equivalently, using stable implementations of M_0 and M_1 ,

$$M_2(U; r) = \frac{a^2}{q_r^2} M_0(U; r) + \frac{\sigma^2}{q_r^2} M_1(U; r) - L(r) \frac{2a\sigma\sqrt{U}}{q_r^2} \phi(d_-).$$

The infinite-window limit is

$$M_2(\infty; r) = L(r) \left(\frac{a^2}{q_r^2} + \frac{a\sigma^2}{q_r^3} \right).$$

2 Erlang-2 Local Guess–Kill CDF

Let

$$D = t_0 + X, \quad G \sim \text{Erlang}(2, \lambda_g), \quad K \sim \text{Erlang}(2, \lambda_k),$$

where the Erlang clocks start at raw trial time 0 and the evidence process starts at t_0 . Let

$$r = \lambda_g + \lambda_k, \quad U = T - t_0.$$

The decision-response CDF term is

$$\int_0^U f_X(x) S_G(t_0 + x) S_K(t_0 + x) dx.$$

Since

$$S_G(t) S_K(t) = e^{-rt} (1 + \lambda_g t) (1 + \lambda_k t) = e^{-rt} [1 + rt + \lambda_g \lambda_k t^2],$$

this term equals

$$e^{-rt_0} [a_0 M_0(U; r) + a_1 M_1(U; r) + a_2 M_2(U; r)],$$

where

$$a_0 = 1 + rt_0 + \lambda_g \lambda_k t_0^2, \quad a_1 = r + 2\lambda_g \lambda_k t_0, \quad a_2 = \lambda_g \lambda_k.$$

The guess-response term is split at t_0 . For $t < t_0$, $S_D(t) = 1$ and the integral is elementary. For $t \geq t_0$, integration by parts reduces the polynomial-exponential integrals involving $F_X(t - t_0)$ to elementary boundary terms plus the same M_0 , M_1 , and M_2 .

3 Uniform Start-Point Variability

For ordinary Wald SPV, the distance a is uniform on

$$a \in [a_L, a_H] = [b - A, b].$$

Let

$$S = \sigma \sqrt{U}, \quad q = q_r, \quad \eta_- = \frac{v - q}{\sigma^2}, \quad \eta_+ = \frac{v + q}{\sigma^2}.$$

The SPV second moment is

$$M_2^{\text{SPV}}(U; r) = \frac{1}{A} [\mathcal{P}(a)]_{a_L}^{a_H},$$

with

$$\mathcal{P}(a) = \frac{H_2(\eta_-, qU; a) + H_2(\eta_+, -qU; a)}{q^2} + \frac{\sigma^2}{q^3} [H_1(\eta_-, qU; a) - H_1(\eta_+, -qU; a)] - \frac{2S}{q^2} K_1(\eta_-, qU; a).$$

The primitive helpers are

$$H_n(\eta, c; a) = \int a^n e^{\eta a} \Phi\left(\frac{c - a}{S}\right) da, \quad K_n(\eta, c; a) = \int a^n e^{\eta a} \phi\left(\frac{c - a}{S}\right) da.$$

Define

$$z = \frac{c - a}{S}, \quad h = z + \eta S, \quad E = \exp\left(\eta c + \frac{1}{2} \eta^2 S^2\right), \quad m = c + \eta S^2.$$

Then

$$K_1 = -SE [m\Phi(h) + S\phi(h)],$$

$$K_2 = -SE \left[(m^2 + S^2)\Phi(h) + S(2m - Sh)\phi(h) \right],$$

$$K_3 = -SE \left[(m^3 + 3mS^2)\Phi(h) + S\{3m^2 - 3mSh + S^2(h^2 + 2)\}\phi(h) \right].$$

For $\eta \neq 0$,

$$H_0 = \frac{e^{\eta a}\Phi(z) - E\Phi(h)}{\eta},$$

$$H_1 = \frac{ae^{\eta a}\Phi(z)}{\eta} - \frac{H_0}{\eta} + \frac{K_1}{\eta S},$$

$$H_2 = \frac{a^2 e^{\eta a}\Phi(z)}{\eta} - \frac{2H_1}{\eta} + \frac{K_2}{\eta S}.$$

For $\eta = 0$, use the polynomial primitives

$$H_1(0, c; a) = \frac{a^2}{2}\Phi(z) + \frac{K_2(0, c; a)}{2S},$$

$$H_2(0, c; a) = \frac{a^3}{3}\Phi(z) + \frac{K_3(0, c; a)}{3S}.$$

For $U = \infty$,

$$M_2^{\text{spv}}(\infty; r) = \frac{1}{A} \left[\frac{e^{\eta_- a}}{q^2} \left(\frac{a^2}{\eta_-} - \frac{2a}{\eta_-^2} + \frac{2}{\eta_-^3} \right) + \frac{\sigma^2 e^{\eta_- a}}{q^3} \left(\frac{a}{\eta_-} - \frac{1}{\eta_-^2} \right) \right]_{a_L}^{a_H},$$

with the polynomial limit used when $\eta_- = 0$.

4 GBM Physical Start-Point Variability

For GBM,

$$dY_t = \mu Y_t dt + \sigma Y_t dW_t,$$

the log process has Wald drift

$$v = \mu - \frac{1}{2}\sigma^2.$$

For physical start-point variability,

$$Y_0 \sim U(1, 1 + A),$$

let $z = \log Y_0$ and $a = \log b - z$. Then

$$e^z = be^{-a}, \quad dz = -da,$$

and

$$a_L = \log \left(\frac{b}{1 + A} \right), \quad a_H = \log b.$$

Therefore

$$M_2^{\text{gbm-spv}}(U; r) = \frac{b}{A} \int_{a_L}^{a_H} e^{-a} M_2(U; r, a, v, \sigma) da.$$

This is the same primitive machinery as ordinary SPV, but with shifted exponents

$$\tilde{\eta}_- = \frac{v - q}{\sigma^2} - 1, \quad \tilde{\eta}_+ = \frac{v + q}{\sigma^2} - 1,$$

and bounds $[a_L, a_H]$. Thus

$$M_2^{\text{gbm-spv}}(U; r) = \frac{b}{A} [\mathcal{P}_{\text{gbm}}(a)]_{a_L}^{a_H},$$

where

$$\mathcal{P}_{\text{gbm}}(a) = \frac{H_2(\tilde{\eta}_-, qU; a) + H_2(\tilde{\eta}_+, -qU; a)}{q^2} + \frac{\sigma^2}{q^3} [H_1(\tilde{\eta}_-, qU; a) - H_1(\tilde{\eta}_+, -qU; a)] - \frac{2S}{q^2} K_1(\tilde{\eta}_-, qU; a).$$

The infinite-window expression is obtained from the ordinary SPV infinite expression by replacing η_- with $\tilde{\eta}_-$, changing the bounds to $[a_L, a_H]$, and multiplying by b/A .

5 Implementation Status

The code uses this closed-form M_2 for Erlang-2 local guess-kill when $sv = 0$, including ordinary Wald SPV and GBM physical-SPV. The existing stable killed-CDF paths supply M_0 and M_1 ; the primitive formula supplies the missing M_2 . Drift variability $sv > 0$ still uses the existing GL integration because it requires the drift-mixture of these moments.